

Enterprise GIS Program Management Plan

A practical framework for organizing enterprise geospatial delivery

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Sanitized public edition

Purpose

This plan translates an enterprise GIS strategy into an operating program. It defines scope, governance, workstreams, delivery controls, risks, quality practices, and change-management responsibilities.

What this sample demonstrates

- Program scope and intended benefits
- Decision authority and team responsibilities
- Workstreams, milestones, and deliverables
- Risk, quality, communication, and change controls
- Performance reporting and continuous improvement

This sanitized edition removes organization names, locations, customer identities, infrastructure details, branch names, approval positions, and funding arrangements while preserving the management approach.

Scope Goals and Benefits

The program establishes and sustains an enterprise GIS capability that supports established business lines, enables new services, and aligns data and technology investments with organizational priorities.

Goals

- Provide easy and reliable access to relevant geospatial information.
- Improve project execution through reusable data, services, and applications.
- Develop standard schemas, policies, procedures, and training.
- Create customer-facing data management and analysis services where demand supports them.
- Maximize return on investment by prioritizing high-value work with a realistic path to delivery.

Program objectives

- Operate enterprise infrastructure for data, services, and applications.
- Establish authoritative data-management and governance practices.
- Develop a repeatable portfolio of GIS products and services.
- Coordinate program delivery across technical specialists, project teams, leadership, and customers.

Intended benefits

The program reduces duplicated effort, makes organizational information more discoverable, improves consistency, supports broader analysis, and provides a controlled path for expanding GIS services.

Selecting Investments

Potential capabilities should be evaluated according to expected business value and ease of launch. This prevents the program from committing resources to attractive ideas that cannot be implemented or sustained.

Technology acquisition strategy

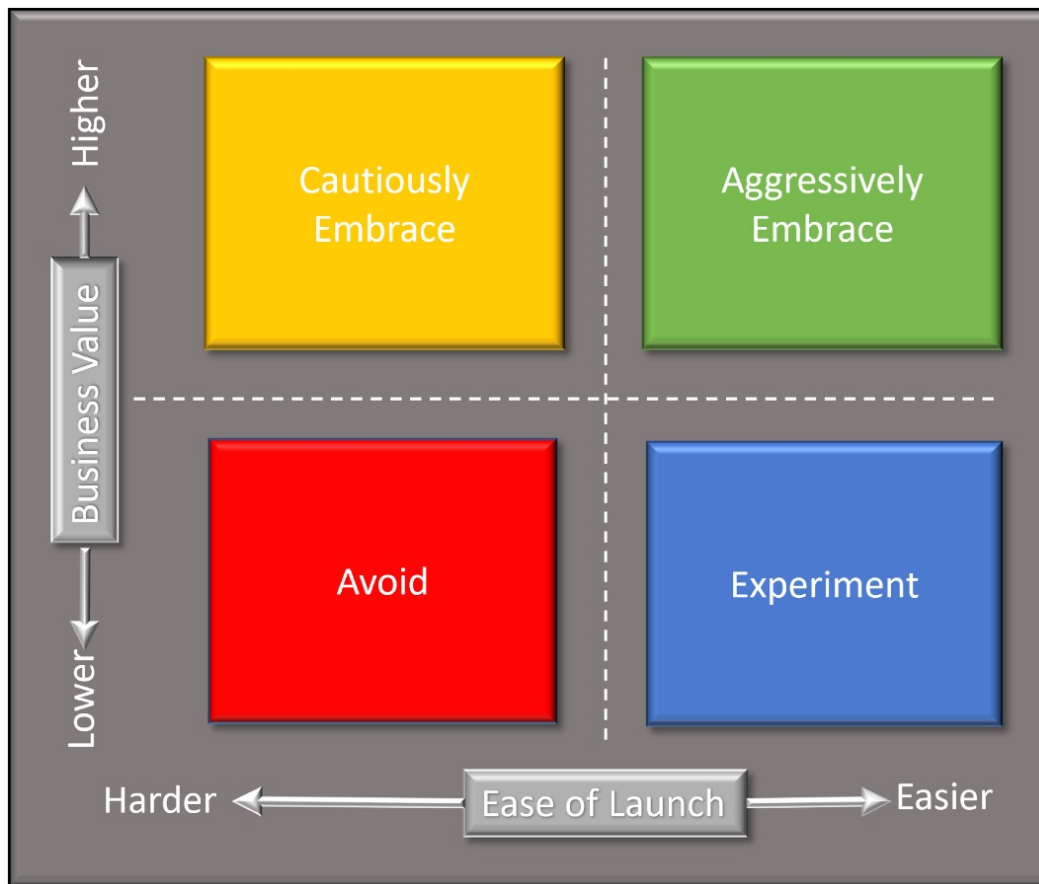


Figure 1. The portfolio matrix supports transparent decisions about which capabilities to adopt, test, defer, or avoid.

Decision criteria

- Expected mission or business value
- Implementation effort and technical readiness
- Availability and quality of required data
- Stakeholder ownership and adoption commitment
- Recurring support, maintenance, and lifecycle cost
- Risk, dependencies, and ability to scale

The program manager records the basis for each portfolio decision and revisits it when requirements, resources, or organizational readiness change.

Program Organization and Decision Rights

Role	Primary responsibilities
Executive sponsor	Confirms strategic alignment, resolves major barriers, approves significant scope or funding decisions, and reviews performance.
Program manager	Directs the program, maintains the plan and roadmap, coordinates resources, manages risks and changes, and reports progress.
GIS program team	Implements architecture, standards, data management, applications, quality controls, training, and technical support.
Data owners and stewards	Approve authoritative sources, define business rules, review quality, and sustain assigned datasets.
Project and business representatives	Define requirements, participate in testing, accept deliverables, and provide adoption feedback.
Technical support partners	Support approved hosting, identity, database, integration, security, and operational requirements.

Decision model

Leadership approves strategy, major investments, and material scope changes. The program manager controls the approved roadmap and coordinates delivery. Data owners decide business meaning and authoritative status. Technical specialists manage implementation within approved standards and access controls.

Communication

The team uses scheduled working sessions for delivery coordination and periodic leadership reviews for decisions, risk escalation, resource alignment, and performance reporting.

Workstreams Milestones and Deliverables

Workstream	Representative deliverables	Completion evidence
Program management	Charter, roadmap, schedule, resource plan, decision log, performance reports	Approved direction, current plans, resolved decisions, documented status
Governance and standards	Policies, roles, schemas, metadata rules, access model, lifecycle procedures	Assigned accountability and consistently applied controls
Authoritative data	Inventory, data dictionaries, quality assessments, migrations, sustained feeds	Trusted datasets with documented sources and update processes
Enterprise platform	Content architecture, services, access page, groups, permissions, operating procedures	Reliable access for authorized users and maintainable administration
Applications and services	Requirement records, prototypes, testing, releases, training, support material	Accepted products tied to defined business needs
Adoption and workforce	Role-based training, communications, staffing plans, support model	Active use, adequate support capacity, and reduced dependence on individual staff

Milestone control

Each workstream moves through documented requirements, design, review, testing, acceptance, publication, and transition to operations. The program manager maintains dependencies and prevents release when essential ownership, security, quality, or maintenance conditions remain unresolved.

Risk and Quality Management

Risk	Potential effect	Management response
Constrained funding or staffing	Priority work cannot be completed or sustained.	Sequence work by business value, forecast capacity, define minimum operating needs, and maintain surge-support options.
Weak stakeholder participation	Products do not fit workflows and adoption remains low.	Confirm ownership, involve users in requirements and testing, and track acceptance and use.
Inconsistent schemas or standards	Data cannot support combined analysis or repeatable applications.	Apply acceptance criteria, validation rules, stewardship review, and controlled schema changes.
Poor archived data or metadata	Legacy holdings cannot qualify as authoritative.	Document provenance, assess fitness, prioritize remediation, and retain limitations where correction is impractical.
Underfunded maintenance	Applications and datasets become outdated.	Assign operational owners, define recurring tasks, and include lifecycle cost in investment decisions.
Technology or integration constraints	Planned services cannot operate reliably.	Engage technical partners early, test dependencies, and use approved architectures and release procedures.

Quality management

Quality begins with documented requirements and continues through iterative review, validation, testing, acceptance, and operational monitoring. Reviews address spatial and thematic accuracy, completeness, consistency, duplication, metadata, accessibility, and fitness for the intended use.

Change Control Reporting and Continuous Improvement

Change control

The program manager evaluates proposed changes for effects on scope, schedule, cost, architecture, data quality, access, staffing, and maintenance. Material changes require the appropriate approval and updates to affected plans. Minor administrative changes remain documented in the change log.

Performance reporting

- Delivery status against approved milestones
- Current risks, decisions, dependencies, and corrective actions
- Dataset quality, stewardship, and update status
- Service availability, use, support demand, and adoption
- Workload, staffing capacity, spending, and forecast needs
- Benefits realized through reuse, efficiency, decision support, and customer value

Operational transition

A deliverable enters routine operations only after the program confirms ownership, maintenance procedures, permissions, support responsibilities, documentation, training, and acceptance. Temporary project teams should not leave unsupported services behind.

Continuous improvement

The team conducts periodic reviews after major releases and at planned program intervals. Lessons learned, performance results, user feedback, and changing business needs inform the next roadmap update.

Evidence of a well managed program

- Current strategy, roadmap, and priorities connect program work to business needs.
- Roles, requirements, milestones, acceptance decisions, and operational transitions remain documented.
- Risks, quality findings, scope changes, and decisions have owners and resolution records.
- Leadership can relate cost and capacity to adoption, reuse, efficiency, customer outcomes, and service demand.

The plan remains a living control document. It provides enough structure to maintain accountability without replacing the working schedules, decision records, technical standards, and operating procedures used by the delivery team.